Teja Ravi
Sr. Data Engineer

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Professional Summary

Experienced **Azure Data Engineer** with 10+ years in strong understanding of Azure data services and solutions. Proficient in designing, building, and maintaining scalable data pipelines within the Azure ecosystem. Skilled in data modelling, ETL processes, and executing data workflow orchestration.

- Experience managing databases and **Azure Data Platform services**, like: **Azure Data Lake (ADLS), Data Factory (ADF)Data Lake Analytics Stream Analytics Azure SQL HDInsight/Databricks**, **NoSQL databases (e.g., HBase, Cassandra, MongoDB), SQL Server, Oracle, Data Warehouse**, successfully built multiple Data Lakes.
- Hands-on experience with Databricks, including: **Databricks Workspace User Interface**, **Managing Databricks Notebooks**, **Delta Lake with Python**, **Delta Lake with Spark SQL**.
- Proficient in working with the Spark ecosystem: **Spark-SQL**, **Spark Core**, **Data frames**, Real-time streaming, **Scala queries**, Handling various data file formats **(Parquet, ORC, Sequence, Text, CSV).**
- In-depth understanding of Snowflake Database: Schema and table structures, experience with **Snowflake Clone** and Time Travel features.
- Successfully migrated existing SQL data and reporting feeds to **Hadoop**.
- Skilled in system analysis, **E-R/dimensional data modelling**, and database design, Hands-on experience with ETL processes using Informatica.
- Extensive experience in text analytics, Proficient in generating data visualizations using **R, SAS, Python,** created dashboards using tools like **Tableau** and **Power BI**.
- Hands-on experience in test-driven development (TDD), behaviour-driven development (BDD), and acceptance test-driven development (ATDD).
- Generated capacity planning reports using Python packages **(NumPy, matplotlib, pandas).**
- Designed and built modern data pipelines and data flows using **Azure Data Factory V2**.
- Designed Azure data storage solutions, including: **SQL Database, SQL Data Warehouse, Data Lake Storage.**
- Hands-on experience in data ingestion, processing, and optimization using **Databricks, Data Lake, and Data Factory**.
- Involved in various phases of data warehouse/data lake projects: **Requirement analysis, ETL development, Performance tuning, Data validation**.
- Leveraged **Azure DevOps** and **Azure Pipelines** for automated builds, releases, and early bug detection, utilized **Azure DevOps** to automate deployment processes for both banking and healthcare projects.
- Integrated **Azure Databricks** with other **Azure services** (e.g., **Azure Data Lake Storage, Azure Event Hub, Azure SQL Database**), demonstrated expertise in custom coding within Azure Logic Apps.
- Good understanding of **NoSQL** databases and hands-on experience in writing applications on platforms like **HBase**, **Cassandra**, and **MongoDB**.

- Experience using **SQOOP** for importing and exporting data from **RDBMS** to **HDFS** and **Hive**.
- Diverse experience across all phases of the software development life cycle (SDLC).
- Worked with Azure Power BI for data visualization.

Technical Skills

Azure Technologies:	Azure Data Lake, Azure Data Bricks, Blob storage, Synapse,Data Storage Explorer, Azure Active Directory, HDInsight,Cosmos DB.
Hadoop Big Data Technologies:	Airflow, Hadoop, Spark, Sqoop, Hive, HBase, Oozie, Flume and Kafka
ETL/Reporting:	Power BI, Data Studio, Tableau
Cloud based Technologies:	Microsoft Azure.
Databases:	MySQL, Oracle, MS-SQL SERVER, Azure SQL Database,Amazon RDS and Google Cloud SQL.
NoSQL Databases:	MongoDB, Cassandra
Programming Languages:	Python, PySpark, Scala, PowerShell
Presentation tools:	MS – Visio, Power Point.

Certifications

Microsoft Certified: Azure Data Engineer Associate

Work Experience

Sr. Data Engineer

Client: Anya Healthcare **June 2023- Present**

Experienced Sr. Data Engineer at Anya Healthcare, responsible for creating a robust and efficient data pipeline to handle electronic health record (EHR)/ EMR data. Successfully stored patient-related data securely and enabled efficient querying. Seamlessly transitioned data from legacy systems to Azure. Safeguarded critical healthcare data against loss or system failures. Key responsibilities included designing scalable data pipelines using Azure Data Factory, defining data ingestion processes, transforming and cleansing data, and collaborating with clinical teams to ensure accurate reporting and analytics.

Responsibilities:

- Designed a scalable data pipeline using Azure Data Factory, leveraged Azure Data Factory to securely ingest and transform EHR (Electronic Health record) data from various sources (e.g., hospitals, clinics, wearable devices), ensured seamless integration with Azure Data Lake Store and Azure Blob storage.
- Utilized Azure Data Factory for data migration in advanced analytics projects, reformed existing ETL processes, transitioning from SQL Server Integration Services (SSIS) to Azure Data Factory.

- Proficient in managing database systems, particularly with a strong focus on healthcare Electronic Medical Record (EMR) systems.
- Experienced in designing, implementing, and maintaining databases to support critical healthcare workflows.
- Scheduled and automated data pipeline execution, monitored pipeline performance and troubleshooted issues.
- Leveraged Azure SQL Database for storing patient demographics, medical history, and treatment records.
- Designed appropriate database schema and tables, implemented data partitioning and indexing strategies for optimal query performance.
- Collaborated with clinical teams to define data access requirements, Wrote T-SQL queries to retrieve relevant information for reporting and analytics.
- Collaborated with clinical stakeholders to understand legacy data structures.
- Worked on Mapping data fields from legacy systems to Azure SQL Database, ensured data consistency and accuracy during migration, implemented data validation checks to identify discrepancies.
- Monitored the migration process and addressed any issues promptly, have Set up automated backup routines for Azure SQL Database, Defined retention policies for backups.
- Worked on Testing data restoration procedures, implemented disaster recovery plans, including failover to secondary regions, regularly validated backup integrity.
- Demonstrated success in efficiently migrating data from legacy systems to new Software-as-a-Service (SaaS) platforms.
- Integrated Azure Data Lake Storage Gen2 to handle large-scale medical imaging data (X-rays, MRIs, CT scans).
- Utilized Azure Databricks for processing unstructured medical data (e.g., natural language processing for clinical notes).
- Optimized data partitioning and indexing within the analytics platform.
- Designed and developed highly efficient data pipelines using Databricks to ingest, store, and process healthcare data from multiple sources, achieved a significant reduction in data processing time, resulting in faster insights and improved operational efficiency.
- Implemented data quality and governance processes within Databricks, ensured compliance with industry regulations, minimized data errors, and enhanced overall data accuracy.
- Successfully migrated data engineering workflows from Cloudera to Databricks, adapted existing ETL processes, ensuring seamless data integration and minimal disruption during the transition.
- Proficiently utilized Kafka for real-time data streaming, integrated Kafka topics into Databricks pipelines, enabling timely processing and analysis of healthcare data.
- Collaborated with data scientists and clinicians to define analytical requirements.
- Worked on Profile query workloads to identify bottlenecks.
- Optimized SQL queries by rewriting, indexing, or caching, monitored resource utilization (CPU, memory, storage) within Azure SQL Database, tuned data partitioning strategies based on query patterns.
- Collaborated with researchers to create custom views or materialized views.

Data engineer

Client: Lloyds Banking Group **Feb 2019- May 2023**

As an ETL Developer at Lloyds Banking Group, I played a pivotal role in managing data lakes using Azure technologies. My responsibilities included integrating data from various sources, optimizing query performance, and ensuring seamless data movement. I successfully migrated on-premises SQL databases to Azure SQL Database, implemented automated backup solutions, and collaborated with business analysts to enhance reporting and analytics. Proficient in SQL, Python, and Scala, I contributed to end-to-end data solutions and maintained data integrity throughout the process.

Responsibilities:

- Worked on design and implementation of a data warehouse solution using Azure Synapse Analytics (formerly SQL Data Warehouse), expertise in handling large-scale data processing and analytics.
- Designed and orchestrated end-to-end data integration workflows using Azure Data Factory.
- Streamlined Extract, Transform, Load (ETL) processes for seamless data movement and transformation.
- Leveraged ADF to ingest data from legacy banking systems, ensuring data consistency and compliance, migrated critical financial data to modern cloud environments and optimized query performance.
- Created complex ETL pipelines using Azure Data Factory in crucial times for efficient transforming and loading data from diverse sources, worked on End-to-End Data Pipeline with Azure Data Lake Storage Gen2.
- Designed and implemented an end-to-end data pipeline using Azure Data Lake Storage Gen2 in managing data lakes.
- Using Data Lake Storage Gen2 performed scalability, security, and flexibility for storing and analysing large volumes of data.
- Developed custom data ingestion scripts in Python for extracting data from various sources, Used Python language for data manipulation and integration.
- Migration to Azure SQL Database:
- Led the migration of on-premises SQL databases to Azure SQL Database and ensured data consistency, security, and performance during the transition, expertise in database management and cloud migration.
- Implemented automated backup and disaster recovery solutions in critical conditions for maintaining data integrity and availability, and focused on data protection and continuity.
- For decision-making, used data modeling concepts to enhance business intelligence reporting and improve data analytics, resulting in actionable insights and strategic planning.
- Worked on Analyzing, designing, and creating modern data solutions using Azure PaaS services with industry trends.
- Engaged in leveraging services such as Azure Data Factory, T-SQL, Spark SQL, and U-SQL to adapt evolving technologies.
- Designed robust CI/CD pipelines to ensure thorough testing and seamless deployment of new features, Leveraged Azure DevOps and Azure Pipelines for automated builds, releases, and early bug detection.

- Orchestrated rigorous testing in the applications using Azure DevOps, conducted functional tests to verify behavior and non-functional tests for performance, security, and scalability.
- Integrated Azure DevOps with Git for effective version control, streamlined code management and collaboration across development teams.
- Created Spark applications using Pyspark and Spark-SQL with Big data processing knowledge.
- Optimized Spark applications by adjusting parameters and monitored clusters ensured efficient execution.
- Migrated data from on-premises sources (e.g., SQL Server or MongoDB) to Azure Data Lake Store (ADLS) for data movement and transformation.
- Worked on data centralization and scalability.

Environment: Azure PaaS Service, Azure data storage, Azure Data Factory (ADF V1/V2), T-SQL, Spark-SQL, U-SQL, Azure SQL, Azure Data Lake Storage (ADLS), Azure DW, Azure Databricks, Blob Storage, PySpark, Scala, Spark Databricks, SQL, Liquibase, JDBC, SQL scripts.

ETL Developer

Client: DHL **Aug 2016- Jan 2019**

As an ETL Developer at DHL, I focused on implementing and maintaining data workflows using Azure Synapse Analytics. My expertise included handling large-scale data processing, optimizing query performance, and ensuring data consistency during migration. I developed and maintained ETL pipelines using Talend, collaborated with cross-functional teams, and provided production support. Proficient in SQL, Python, and Scala, I contributed to critical ETL projects from inception to deployment, emphasizing documentation, best practices, and effective problem-solving.

Responsibilities:

- Collaborated with cross-functional teams to develop and maintain ETL workflows using Azure Data Factory.
- Ensured data consistency, security, and performance during the migration of on-premises SQL databases to Azure SQL Database.
- Implemented automated backup and disaster recovery solutions.
- Developed and maintained ETL pipelines using tools such as Talend, ensuring seamless data movement between sources.
- Proficient in SQL, Python, and Scala for data extraction, transformation, and loading.
- Worked with internal teams to optimize data workflows and improve efficiency, conducted thorough quality assurance tests on ETL processes to validate data accuracy.
- Debugged and resolved issues promptly, minimizing production downtime, implemented data validation checks and error handling mechanisms.
- Tuned ETL processes for optimal performance, including query optimization and indexing,
- Improved data processing efficiency through performance enhancements.
- Monitored data pipelines and proactively addressed bottlenecks.
- Worked with data warehouses (e.g., Azure Synapse Analytics, Snowflake) to handle large volumes of data.
- Collaborated with business analysts to translate requirements into effective data structures.

- Worked on ETL projects from inception to deployment, ensuring timely delivery, documented ETL processes, workflows, and best practices.
- Provided production support for critical ETL processes, resolving incidents promptly,
- Managed change requests, version control, and deployment processes.
- Collaborated with business users to understand their needs and adapt ETL solutions accordingly.

ETL Production Support

Client: Open origin **March 2013- Jul 2016**

In my role as ETL Production Support at Open Origin, I provided critical production support for ETL processes. My responsibilities included investigating and resolving data-related issues promptly, collaborating with cross-functional teams, and interacting with vendors. I managed daily workloads, documented solutions, and ensured system stability. Proficient in monitoring performance metrics and providing after-hours support, I contributed to maintaining data accuracy and minimizing downtime. My focus on problem-solving and collaboration allowed me to address critical issues effectively.

Responsibilities:

- Provided production support for critical ETL processes, ensuring data accuracy and reliability,
- Investigated and resolved data-related issues promptly, minimizing downtime.
- Analyzed and resolved production issues promptly, ensuring minimal downtime, joined bridge lines and provided timely updates during critical incidents.
- Collaborated with cross-functional teams to troubleshoot ETL processes.
- Interacted with vendors to address application-related concerns.
- Managed daily workload based on priorities and maintained SLAs, provided support to end users, addressing inquiries and concerns.
- Documented solutions and created a knowledge base for effective problem-solving, developed technical documentation and presented findings to management.
- Prepared artifacts for future development and troubleshooting.
- Analyzed monitor calibration reports and workstation logs, Identified and resolved calibration errors to maintain data accuracy.
- Monitored performance metrics and ensured system stability.
- Provided after-hours on-call production support, assisted with complex desktop setups, including multiple monitors and peripherals.
- Collaborated with team members to address critical issues.

Education

Bachelors in Computer Science, Graduated: May 2012